

Quantum Field Theory and Statistical Systems

Discrete microstate spectra and Schottky anomaly in charged de Sitter black holes via bumblebee gravity

Nazek Alessa ^a, M. Umair Shahzad ^{b,c,*}, Anees Ashfaq ^b^a Department of Mathematical Sciences, College of Science, Princess Nourah bint Abdulrahman University, P.O. Box 84428, Riyadh 11671, Saudi Arabia^b Department of Mathematics, Faculty of Sciences, University of Okara, Okara 56130, Pakistan^c International Center for Materials Sciences and Technology, Western Caspian University, Baku, AZ 1001, Azerbaijan

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ABSTRACT

This paper is devoted to study the thermodynamics of charged de Sitter black holes in Einstein–Bumblebee gravity, a Lorentz-violating extension of general relativity, using an effective two-horizon framework that treats the region between the event and cosmological horizons as a single thermodynamic system. We derive globally consistent thermodynamic variables such as effective temperature, pressure, and volume and show that they satisfy a generalized first law. The effective heat capacity shows a strong Schottky-type anomaly, which is defined by a single-peak structure resulting from the interaction between the two horizons. It is demonstrated here that Lorentz-violating effects cause this feature to undergo systematic and controlled deformations. In particular, we observe that the Lorentz-violation parameter leads to a displacement and rescaling of the heat-capacity peak, which can be interpreted as a change of the effective excitation scale for the thermodynamic response. We also show that the overall thermodynamics of the two-horizon system can be effectively described by a phenomenological finite-level model, where the temperature difference between the horizons acts as an effective excitation gap. Although the map is not derived microscopically, it accurately reproduces the shape of the heat-capacity curve and allows one to interpret the macroscopic thermodynamical enhancement. Our results reveal that Lorentz-violating corrections act like a controlled change to the thermodynamics of black holes with multiple horizons. They keep the overall Schottky-type pattern but change the specific numerical details. This highlights the strong sensitivity of horizon thermodynamics to corrections beyond standard gravity and provides a systematic approach to study these deviations within a semiclassical framework.

1. Introduction

Black hole (BH) thermodynamics provides one of the most profound conceptual bridges linking gravity, statistical physics, and quantum mechanics. The four laws of BH mechanics [1,2] together with the prediction of Hawking radiation [3] demonstrated that BHs behave as genuine thermodynamic systems characterized by entropy, temperature, and horizon area. Subsequently, extensive investigations have been conducted on the thermodynamics of BHs in various spacetimes, revealing many new aspects of entropy, phase transitions, and quantum corrections [4–8]. The current research has increasingly focused on multi-horizon spacetimes that extend beyond the Reissner–Nordström–de Sitter (RN–dS) model. The unique temperature characteristics that distinguish between BH

* Corresponding author.

E-mail addresses: mushahzad@uo.edu.pk (M.U. Shahzad), aneeshahfaq2002@gmail.com (A. Ashfaq).

and cosmological boundaries require researchers to study two boundary systems to test their thermodynamic methods against a single central system [9–11]. Recent studies have shown that, under suitable equilibrium conditions, these systems provide heat capacity behavior which resembles phase transitions found in common thermodynamic matter [12–14].

According to the seminal work of Kostelecký and Samuel from 1989, the Bumblebee model exists as a theoretical framework that researchers can use to investigate violations of Lorentz symmetry [15,16]. A vector or tensor field that develops a non-zero vacuum expectation value is used in the string theory-based scenario to create Lorentz violations. The Bumblebee gravity formalism demonstrates a violation through a potential function $V(B_\mu B^\mu)$ which causes the vector field B_μ to undergo spontaneous symmetry breaking [17,18]. The model has been extended to curved backgrounds through Riemannian and Riemann-Cartan geometries [19–25]. A vector field vacuum value that breaks Lorentz symmetry is used in the theory of Einstein-bumblebee gravity to produce new particle interactions and BH solutions that have been thoroughly investigated [15,26,27]. In Lorentz-violating theories, the thermodynamic properties of BHs show variations in stability conditions, entropy products, and Smarr relations [28–32]. The first static BH solutions in bumblebee gravity were obtained by Casana et al. [33], which led to additional research and generalizations in later work [34–39].

Johnson and collaborators [40] demonstrated that certain charged dS BHs display a Schottky-type anomaly, which is a pronounced peak in the heat capacity analogous to that of a two-level quantum system. Similar observations were made by Gomberoff and Teitelboim [41], Murdia et al. [42], Ma et al. [43], Zhen et al. [44,45], Ma and Zhao [46] and others, who emphasized the role of horizon interactions in producing this discrete-level signature. These results suggest that horizon thermodynamics may encode information about a finite number of microscopic degrees of freedom. However, all such studies were performed within standard general relativity, assuming exact local Lorentz symmetry. The formation of the Schottky anomaly is a remarkable thermodynamic property of these systems. According to Pathria [47–50], the Schottky anomaly is a feature of statistical physics that describes a two-level system characterized by a heat capacity that vanishes at low temperatures. Similar irregularities have been observed in BH thermodynamics [13], where multi-horizon spacetimes or discrete-level signatures produce the same heat capacity behavior [51]. The discovery of these anomalies provides a low-level understanding of BH thermodynamics, suggesting that horizon states and particle transitions between horizons can be quantized.

It is essential to highlight that thermodynamics in dS space is inherently subtle because of the absence of global time-like Killing vector fields and the lack of strict global thermal equilibrium. Moreover, in the multi-horizon scenario, the event and the cosmological horizons will generally have different temperatures, which means it is not possible to construct the usual equilibrium thermodynamic ensemble. An effective (or quasi-equilibrium) thermodynamic scheme is used in the present work, in which the space between the two horizons is considered as a composite system with global variables. It is to be emphasized that this scheme does not imply strict equilibrium. Rather, it is a scheme for the consistent phenomenological description of the thermodynamic behavior of the system, which is widely used to describe the thermodynamics of multi-horizon spacetimes [41–46].

In this work, we extend the thermodynamic analysis of RN-dS BHs to the Einstein-Bumblebee gravity framework. We derive the effective temperature T_{eff} , entropy, and heat capacities $C_{Q,L}$ and $\hat{C}_{Q,L}$, and show that the two-horizon system naturally exhibits a Schottky-type anomaly. This anomaly arises because the two horizons can be regarded as distinct energy levels of a quantum system, with transitions between them generating the observed thermodynamic behavior. The existing research has focused primarily on Lorentz-invariant frameworks, which prevents researchers from understanding how these thermodynamic characteristics change in theories that use different spacetime symmetry principles. It is important to note that, while the impact of Lorentz-violating corrections remains uncertain, whether they yield only known effects or give rise to entirely novel phenomena, the influence of these corrections on thermodynamic properties has yet to be fully understood. Addressing this research question requires not only the extension of multi-horizon thermodynamics into new scientific inquiry but also a thorough investigation of how key parameters influence observable thermodynamic properties.

The paper is organized as follows: In Section 2, we review the Einstein–Bumblebee model and the charged dS solution used here. Section 3 derives the effective thermodynamic quantities for the two-horizon system and explains the boundary conditions used. Section 4 presents numerical results of temperature profiles and heat capacity plots, and Section 5 discusses the two-level analogy, including estimates of the gap Δ and particle number N . We conclude in Section 6 with a summary and outlook.

2. Charged de Sitter black in bumblebee gravity

The bumblebee model extends general relativity by incorporating a vector field, termed the *bumblebee field*, which interacts with the gravitational field via a non-minimal coupling constant. Using the appropriate potential, the field B_μ develops a non-zero vacuum expectation value, and triggering spontaneous symmetry breaking and resulting in the violation of Lorentz symmetry in the gravitational sector. The corresponding action of the model is given by Kostelecký [17].

$$S = \int d^4x \sqrt{-g} \left[\frac{1}{2\kappa} (R - 2\Lambda) + \frac{\xi}{2\kappa} B^\mu B^\nu R_{\mu\nu} - \frac{1}{4} B_{\mu\nu} B^{\mu\nu} - V(B_\mu B^\mu \pm b^2) \right] + \int d^4x \sqrt{-g} \mathcal{L}_M, \quad (1)$$

Here, Λ denotes the cosmological constant, $\kappa = \frac{8\pi G}{c^4}$ is the gravitational coupling constant, and ξ represents the non-minimal coupling constant that governs the interaction between gravity and the bumblebee field.

In the previous study conducted by Casana et al. [33] an exact BH solution was constructed in the absence of a cosmological constant and the matter field A^μ . Building upon this, a charged BH solution with a cosmological constant in the bumblebee gravity framework has been discussed in Liu et al. [52]. The corresponding line element of static and spherically symmetric spacetime is

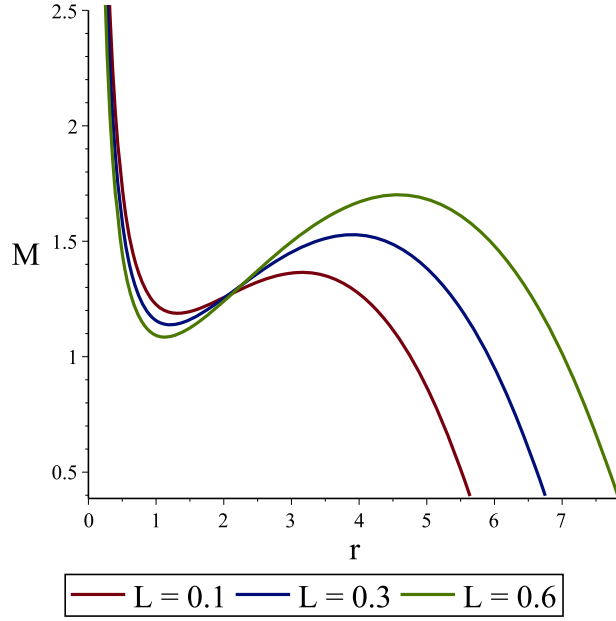


Fig. 1. M - r curve by fixing $Q = 1$ and $l^2 = 100$.

given by

$$ds^2 = A(r)dt^2 - \frac{1+L}{A(r)}dr^2 - r^2(d\theta^2 + \sin^2\theta d\phi^2), \quad (2)$$

where

$$A(r) = 1 - \frac{2M}{r} + \frac{2(1+L)}{2+L} \frac{Q^2}{r^2} - \frac{1}{3}(1+L)r^2\Lambda. \quad (3)$$

The metric function describes the spacetime in terms of mass M , charge Q , cosmological constant Λ , and a Lorentz violation parameter L . This modification of the fundamental geometric structure of spacetime has profound implications for thermodynamics. The nature of the cosmological background is determined by the sign of Λ : the solution corresponds to an asymptotically dS BH for $\Lambda > 0$ and anti-de Sitter (AdS) BH for $\Lambda < 0$. We note that the standard RN-dS/AdS solution is recovered in the limit $L \rightarrow 0$, confirming that L parameterizes pure Lorentz-violating deviations from general relativity. Moreover, it is convenient to express the cosmological scale in terms of the dS radius l , by substituting

$$\Lambda = 3/l^2,$$

in Eq. (3) and we get

$$A(r) = 1 - \frac{2M}{r} + \frac{2(1+L)}{2+L} \frac{Q^2}{r^2} - \frac{(1+L)r^2}{l^2}. \quad (4)$$

The equation $A(r_{+,c}) = 0$ is satisfied by BH and the cosmological horizons at r_+ and r_c , respectively. Inserting Eq. (4), we have

$$M = \frac{r}{2} \left(1 + \frac{2(1+L)}{2+L} \frac{Q^2}{r^2} - \frac{(1+L)r^2}{l^2} \right), \quad (5)$$

Fig. 1 depicts the mass M of the BH as a function of its horizon radius r for distinct values of the Lorentz-violating parameter L in Einstein-Bumblebee gravity. The characteristic swallowtail structure of the $M(r)$ curves confirms a rich phase behavior. For all values of L , the mass-radius relation exhibits a pronounced minimum-maximum profile. This non-monotonic behavior is a signature of a first-order phase transition between thermodynamically stable and unstable BH configurations. The Lorentz-violating parameter L plays a systematic role in this profile. In other words, increasing L parameter will shift the critical point to a greater horizon radius and mass. This means that Lorentz violation will expand the range of existence for BH solutions and increase the maximum possible mass for a stable configuration.

In Fig. 2, the mass horizon relation $M(r)$ is plotted, which describes the thermodynamic phase structure of the BH. The swallowtail feature in the mass-radius curve is a characteristic sign of a first-order phase transition in the BH system. The local minimum A and maximum C are the critical points. It is not possible to have a stable equilibrium state because the states along this curve, which connects A and C , are thermodynamically unstable and have negative heat capacity. The critical mass M_C is the maximum mass for a stable BH, which is obtained from the maximum at C . The mass of this size intersects the curve at three points, which means that

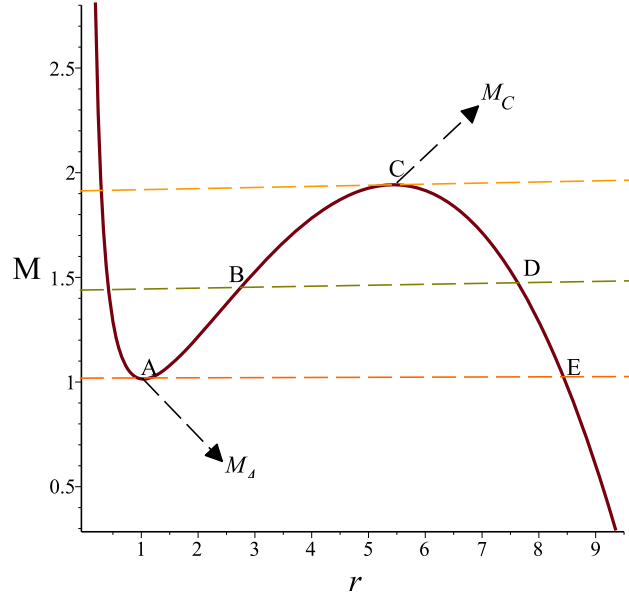


Fig. 2. The behaviour of M as a function of r . Setting $Q = 1$, $l^2 = 100$, $M_A = 1.035$ and $M_C = 1.936$.

there are two phases of BH, small and large, between which a phase transition is possible. This is the gravitational equivalent of the liquid-gas phase transition. The line D , which intersects the unstable branch of the curve at masses greater than the critical mass, confirms the fact that BH solutions are unstable for masses greater than the critical mass [53].

The radii r_A and r_C are found by solving the system of equations consisting of the horizon condition $A(r) = 0$ and the extremality condition $\partial A(r)/\partial r = 0$. Solving these simultaneously gives the expression for the critical radii:

$$r_{A,C}^2 = \frac{l^2}{6} \left(1 \pm \sqrt{1 - \frac{24Q^2(1+L)^2}{(2+L)l^2}} \right). \quad (6)$$

The radius r_A is related to the negative branch of the solution and corresponds to the minimum mass M_A . The radius r_C from the positive branch is associated with the maximum mass M_C . These critical radii establish the stability intervals of the BH solution. The real critical radii r_A and r_C exist provided the parameter bound is satisfied i.e. $\frac{24Q^2(1+L)^2}{(2+L)l^2} \leq 1$. Substituting r_A and r_C in Eq. (5) gives M_A and M_C respectively:

$$M_A = \frac{l^2(1+L)^{1/2} \left(24Q^2(1+L)^2 + \left(1 + \sqrt{1 - \frac{24Q^2(1+L)^2}{l^2(2+L)}} \right) \right)}{3\sqrt{6} \sqrt{l^2 \left(1 + \sqrt{1 - \frac{24Q^2(1+L)^2}{l^2(2+L)}} \right)}}, \quad (7)$$

$$M_C = \frac{l^2(1+L)^{1/2} \left(24Q^2(1+L)^2 - \left(1 + \sqrt{1 - \frac{24Q^2(1+L)^2}{l^2(2+L)}} \right) \right)}{3\sqrt{6} \sqrt{l^2 \left(1 + \sqrt{1 - \frac{24Q^2(1+L)^2}{l^2(2+L)}} \right)}}. \quad (8)$$

These critical masses give a lower and an upper mass limit for stable BH solutions in parameter space. These formulas facilitate the study of the phase structure and stability boundaries. Furthermore, it can be demonstrated that when the event horizon and the cosmological horizon are treated as separate thermodynamic systems, the quantities defined on each individually satisfy the first law of thermodynamics. This separate consistency underscores the robustness of the thermodynamic formalism applied to this Lorentz-violating spacetime.

The thermodynamic quantities related to the event horizon include:

$$dM = T_+ dS_+ + V_+ dP + \varphi_+ dQ \quad (9)$$

with

$$T_+ = \frac{A'(r_+)}{4\pi} = \frac{1}{4\pi r_+} \left(1 - \frac{3r_+^2}{l^2} - \frac{2Q^2(1+L)}{r_+^2(2+L)} \right),$$

$$S_+ = \pi r_+^2, \quad \varphi_+ = \frac{2Q(1+L)}{r_+(2+L)}. \quad (10)$$

The thermodynamic quantities related to the cosmological horizon include:

$$dM = -T_c dS_c + V_c dP + \varphi_c dQ \quad (11)$$

with

$$T_c = -\frac{A'(r_c)}{4\pi} = -\frac{1}{4\pi r_c} \left(1 - \frac{3r_c^2}{l^2} - \frac{2Q^2(1+L)}{r_c^2(2+L)} \right),$$

$$S_c = \pi r_c^2, \quad \varphi_c = \frac{2Q(1+L)}{r_c(2+L)}. \quad (12)$$

We note the conventional sign choice in defining the Hawking temperatures for the event and cosmological horizons. The surface gravity at the cosmological horizon is opposite in sign to the event horizon; consequently, the expression obtained from $T_c = -A'(r_c)/(4\pi)$ carries an explicit minus sign. Physical temperatures are positive definite, and one may define $T_c^{\text{phys}} = |T_c|$. The negative sign simply reflects the outward-pointing normal convention used for the cosmological horizon. For thermodynamic constructions, we use the magnitudes of the horizon temperatures while keeping track of sign conventions when deriving the effective first law [10,54]. Equilibrium conditions such as these $T_+ = T_c$ must therefore be understood as equality of the magnitudes of the respective Hawking temperatures.

Moreover, we have

$$M = \frac{r_c x}{2(1+x+x^2)} + \frac{Q^2(1+x^2)(1+L)}{r_c x(1+x+x^2)(2+L)},$$

$$\frac{1}{l^2} = \frac{1}{r_c^2(1+x+x^2)(1+L)} - \frac{2Q^2}{r_c^4 x(1+x+x^2)(2+L)}. \quad (13)$$

The ratio of the event horizon to the cosmological horizon represented by $x = r_+/r_c$. When $l^2 = \zeta \frac{12Q^2}{(1+L)}$, where $\zeta \geq 1$ is a constant, from Eq. (13), we have

$$r_c^2 = \frac{6Q^2\zeta}{(1+L)(1+x+x^2)} \left(1 + \sqrt{1 - \frac{2(1+L)(1+x+x^2)}{3\zeta x(2+L)}} \right). \quad (14)$$

We introduce the dimensionless parameter $\zeta \geq 1$ via $l^2 = \frac{12\zeta Q^2}{1+L}$; ζ parametrizes the relative scale between the cosmological radius and the charge length-scale Q . Choosing $\zeta \geq 1$ places the model in the physically interesting regime where the cosmological scale is not parametrically smaller than the charge scale, and it simplifies the discriminant analysis, ensuring a sizable parameter space with two nondegenerate horizons.

The Hawking temperatures at event and cosmological horizons, T_+ and T_c , are defined in Eqs. (10) and (12). Substituting Eq. (14) into these temperature formulas produces Eqs. (15) and (16), which express the temperatures explicitly in terms of r_c , x , Q , and L . These results are essential for probing the thermal behavior of the BH and will be used to investigate the Schottky anomaly in the heat capacity.

$$T_+ = \frac{(1-x)}{4\pi r_c x(2+L)} \left(\frac{(1+2x)(2+L)}{(1+x+x^2)} - \frac{2Q^2(1+2x+3x^2)(1+L)}{r_c^2 x^2(1+x+x^2)} \right), \quad (15)$$

$$T_c = \frac{(1-x)}{4\pi r_c(2+L)} \left(\frac{(2+x)(2+L)}{(1+x+x^2)} - \frac{2Q^2(3+2x+x^2)(1+L)}{r_c^2 x(1+x+x^2)} \right). \quad (16)$$

The total entropy of spacetime sums the entropies of the two horizons [10,12,55]. Temperature and interaction entropy were introduced by Zhang et al. [56], Ma et al. [57] in the “thermal radiation” scenario by using the two horizons. This forms the criteria for investigating the thermodynamic properties of the dS spaces and for studying the entropic forces between the two horizons. This method assumes that at the end of the two subthermal systems, E , the temperature of the cosmological horizon, T_c , is the only temperature present in spacetime, and the radiation temperature at the event horizon, T_+ , is zero. We consider the effective thermodynamic parameters of electrically charged BHs, accounting for the role of Lorentz violation and cosmological parameters that have been overlooked in previous literature. In particular, we discuss the relationships among horizon radii, the mass of the BH, and thermal parameters, and determine the roles of these parameters in the appearance of the Schottky anomaly. This paper presents a comprehensive thermodynamic study of various boundary conditions, providing new insights into BH thermodynamics of the Lorentz-violation parameter L in gravitational physics. We emphasize that $L \rightarrow 0$, our results reduce to the RN-dS case; in particular, the Schottky-type heat-capacity anomaly discussed here is already present in the RN-dS limit [45], and the role of L is to quantitatively shift the excitation gap and peak position away from that baseline.

3. Effective thermodynamic quantities in charged dS black holes in Einstein–Bumblebee gravity

Before proceeding with the effective thermodynamic variables, let us first note that this construction is understood in a quasi-equilibrium sense. Indeed, in de Sitter spacetime, a thermodynamic equilibrium cannot be defined in a global sense, as there is no global timelike Killing vector available. Therefore, while these thermodynamic variables are not associated with a basic equilibrium system, they represent an effective thermodynamic description of the system in the region between the two horizons. Such representations have been found to be useful in describing qualitatively a variety of thermodynamic properties of systems with multiple horizons.

$$dM = T_{eff} dS - P_{eff} dV + \varphi_{eff} dQ, \quad (17)$$

where the thermodynamic volume between the event and the cosmological horizons is given by Sekiwa [10], McInerney et al. [55], Dolan et al. [58]

$$V = V_c - V_+. \quad (18)$$

It can be observed that when the potential, represented by φ_A , at the event horizon at point A satisfies the required conditions, we have

$$\frac{Q^2}{r_A^2} = \frac{(1+2x)(2+L)}{2(1+2x+3x^2)(1+L)}. \quad (19)$$

The boundary conditions that ensure global thermodynamic consistency: Eq. (19) enforces zero net heat flow between the two horizons, defining a quasi-equilibrium effective temperature T_{eff} , while Eq. (26) follows from the Clausius relation applied to the inter-horizon volume, giving the net mechanical response as the effective pressure P_{eff} .

The radiation temperature at the event horizon is denoted by $T_+ = 0$, while the radiation temperature at the cosmological horizon, corresponding to point E , is

$$T_E = \frac{3(1-x)^2(1+x)^3}{4r_c x \pi (1+2x+3x^2)(1+x+x^2)}. \quad (20)$$

In case the radiation temperature at a point A goes to zero, then the electric potential on the event horizon of the charged dS BHs in bumblebee gravity satisfies the condition in Eq. (19) and the effective temperature should be T_E . Under that condition, the effective temperature T_{eff} can be found using Eqs. (17) and (18) together with the boundary condition given in Eq. (20).

$$T_{eff} = \frac{1-x}{4\pi r_c x^5 (2+L)} \left[(1-x-x^2+2x^3)(2+L) - \frac{2Q^2}{r_c^2 x^2} (1+x-x^3+x^4)(1+L) \right], \quad (21)$$

and P_{eff} , the effective pressure is given by

$$P_{eff} = \frac{1}{2(1+r_c x^2-x^3)} \left[F(x) \left\{ \frac{x^2-1}{2(1+x+x^2)^2} + \frac{Q^2(1+L)}{r_c^2 x^2 (2+L)} \frac{(1+2x+2x^2+x^4)}{(1+x+x^2)^2} \right\} \right. \\ \left. + \frac{x}{4} F'(x) \left\{ \frac{1}{1+x+x^2} - \frac{2Q^2(1+L)}{r_c^2 x^2 (2+L)} \frac{(1+x^2)}{1+x+x^2} \right\} \right], \quad (22)$$

with

$$F(x) = \frac{8}{5}(1-x^3)^{2/3} + \frac{2}{5(1-x^3)} - 1 = 1+x^2+f_0(x), \\ f_0(x) = \frac{8}{5}(1-x^3)^{2/3} - \frac{8+5x^2-10x^3-5x^5}{5(1-x^3)}. \quad (23)$$

The effective thermodynamic system's potential, φ_{eff} , the entropy S , and the thermodynamic volume V , are as follows

$$\varphi_{eff} = \frac{2Q(1+x^2)(1+L)}{r_c x (1+x+x^2)(2+L)}, \quad S = \pi r_c^2 F(x), \quad V = \frac{4}{3} \pi r_c^3 (1-x^3). \quad (24)$$

The effective quantities are constructed by treating the region between the two horizons as the thermodynamic system and imposing a set of boundary conditions that relate the local horizon temperatures and potentials [54,56,57]. The function $F(x)$ collects geometric factors that come from the area sum $S = S_+ + S_c = \pi r_c^2 (1+x^2)$ together with subleading interaction corrections; explicitly, we write

$$S = \pi r_c^2 (1+x^2+f_0(x)) \equiv \pi r_c^2 F(x),$$

where $f_0(x)$ encodes the interaction entropy between horizons. The analytical form chosen for $F(x)$ in Eq. (23) is a compact representation that reproduces the limits of $x \rightarrow 1$ and is motivated by the requirement that $F(x) \rightarrow 1+x^2$ when interactions are negligible. This construction follows the standard effective two-body thermodynamics approach of Pathria [47], Huang [48] where entropies of the composite system are expressed as sums plus controlled corrections due to interactions.

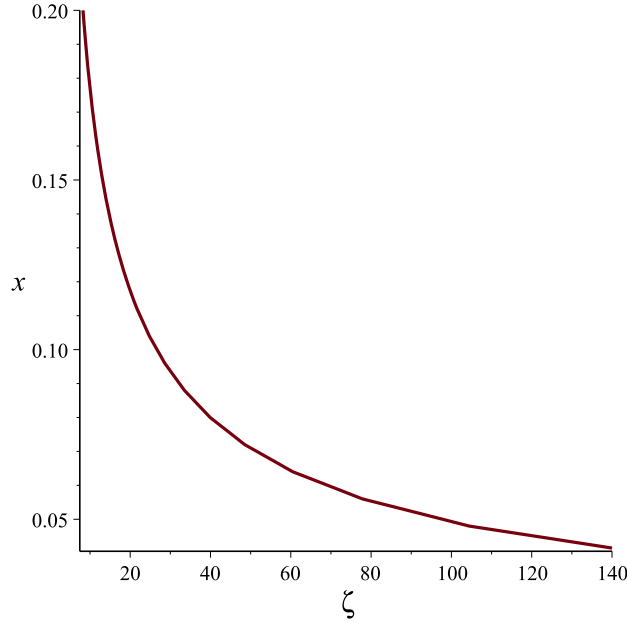


Fig. 3. The $x_{\min}$ - ζ curve by fixing $L = 1$ and $Q = 1$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

From Eqs. (13) and (19), one can find

$$\frac{2(1+2x+3x^2)(1+L)}{x^2(1+2x)(2+L)} = \frac{6\zeta Q^2}{(1+L)(1+x+x^2)} \left(1 + \sqrt{1 - \frac{2(1+x+x^2)(1+L)}{3\beta x(2+L)}} \right). \quad (25)$$

Fig. 3 illustrates how the cosmological scaling parameter ζ affects the minimum horizon ratio $x_{\min}$. This quantity defines the lower bound of the parameter space for which two nondegenerate horizons exist and a consistent thermodynamic description of the system can be developed. As ζ increases and $x_{\min}$ decreases monotonically, the physically admissible configuration space grows toward smaller values of x . This behavior allows a greater range of horizon configurations to satisfy the two-horizon condition, reflecting the growing separation between the characteristic cosmological and charge length scales. It is crucial to distinguish this boundary behavior from the thermodynamic response of the system. The variation of $x_{\min}$ with ζ characterizes the domain of allowed configurations and does not determine the location of extrema in thermodynamic quantities. In particular, it does not imply a corresponding shift of the heat-capacity peak toward smaller values of x . As discussed in Section 4, the position of the Schottky-type maximum in the heat capacity exhibits a different dependence on ζ : when expressed in terms of the geometric variable x , the peak shifts toward larger values of x , whereas in the thermodynamic representation based on the effective temperature T_{eff} , it is displaced toward lower temperatures. This distinction arises from the nontrivial mapping between x and T_{eff} and must be taken into account when interpreting the ζ -dependence of the anomaly.

When the potential of the event horizon satisfies

$$\frac{Q^2}{r_+^2} = \frac{2+L}{2(1+L)(1+x^2)}, \quad (26)$$

the radiation temperature of the event horizon, $\bar{T}_+$, and the cosmological horizon, $\bar{T}_c$, are equivalent.

$$\bar{T}_+ = \bar{T}_c = \frac{1-x}{2\pi r_c(1+x)^2}. \quad (27)$$

Substituting Eq. (26) in Eq. (21), we have

$$\bar{T}_{\text{eff}} = \bar{T}_+ + \frac{(1-x)}{2\pi r_c x^3(1+x)^2} = \bar{T}_c + \frac{(1-x)}{2\pi r_c x^3(1+x)^2}. \quad (28)$$

The radiation temperatures of both horizons now become equal in Eq. (27), establishing thermodynamic equilibrium across spacetime when the electric potential on the event horizon reaches a special equilibrium value in Eq. (26). The effective temperature essentially encapsulates the effect of horizon separation and BH charge, integrating them as a single global measure of the thermality of both horizons. This can be considered beneficial when the different ratios of the horizons change, whereas the effective temperature exhibits a nontrivial thermodynamic behavior, altering the heat capacity peak and the Schottky anomaly. Additionally, the thermal

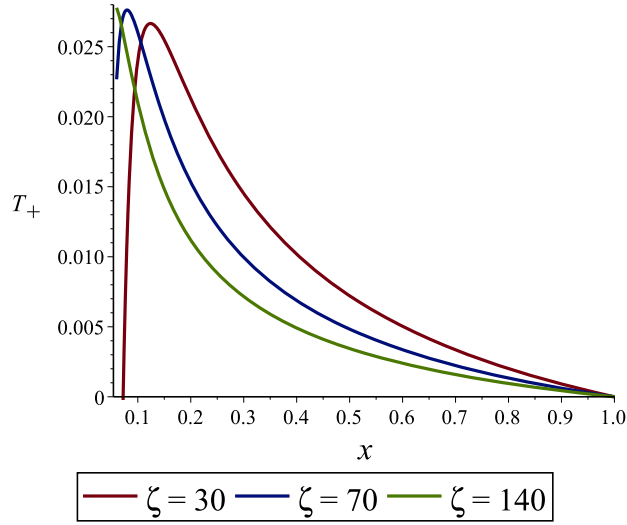


Fig. 4. The T_+ - x curve for different ζ by fixing $L = 0.4$ and $Q = 1$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

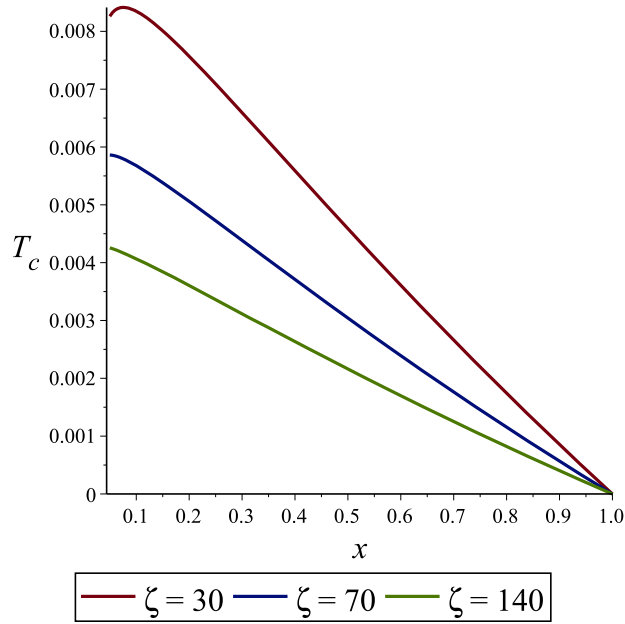


Fig. 5. The T_c - x curve for different ζ by fixing $L = 0.4$ and $Q = 1$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

contribution of horizons is also modified by the Lorentz-violation parameter, which would further alter the location and height of the anomalies. This forms a coherent and physically meaningful manner of studying the global thermodynamic characteristics in multi-horizon spacetimes; it links small microscopic horizon excitations to macroscopic observables and is thus prescribed to stability and equilibrium.

The three temperatures, T_+ , T_c , and T_{eff} are plotted versus the ratio between the two horizon positions, x , for different values of the parameter ζ by using Eqs. (14)–(16) and (21) (taking the positive sign in Eq. (14)).

The thermal behavior of the system, as captured in T_+ , T_c , and T_{eff} , actually constitutes the main mechanism responsible for the Schottky anomaly in this BH model of Lorentz violation. One can observe in Figs. 4–6, the event horizon has a sharp peak in the temperature at low values of x , indicating the maximum thermal radiation emitted by the event horizon. On the other hand, the cosmological horizon has a continuously decreasing temperature with increasing values of x and is suppressed for higher values of ζ .

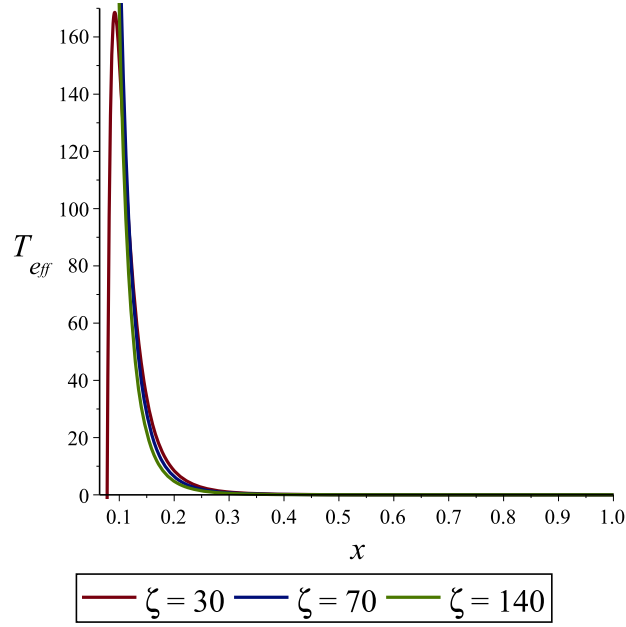


Fig. 6. The T_{eff} - x curve for different ζ by fixing $L = 0.4$ and $Q = 1$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

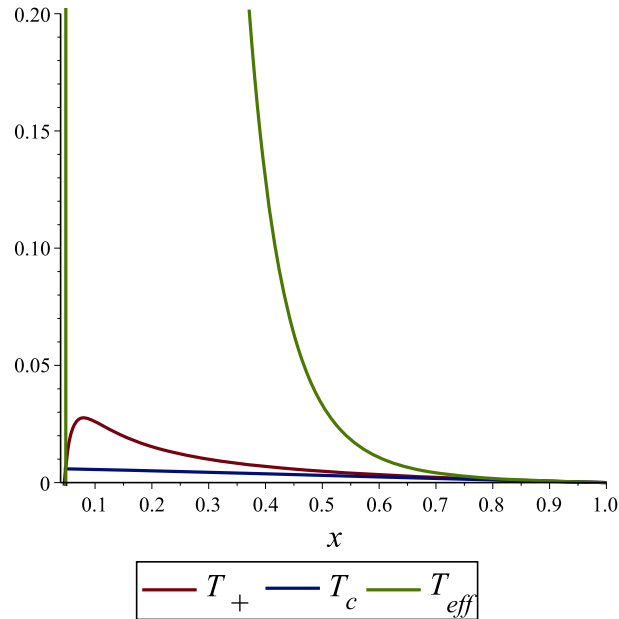


Fig. 7. The T_+ , T_c and T_{eff} curves by fixing $\zeta = 70$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

This indicates the reduction of radiation emitted by the cosmological horizon. In contrast, the effective temperature in Fig. 7 is the dominant factor in the thermal radiation and has a sharp and very narrow peak at low values of x , exceeding the radiation emitted by the event T_+ and cosmological horizons T_c . This sharp peak in the effective temperature is strongly correlated with the non-monotonic peak in the heat capacity, indicating the occurrence of the Schottky anomaly. The sharp rise in the peak of the effective temperature T_{eff} indicates the activation of the horizon's microstates at the maximum point of the peak, and the falling off of the peak with increasing values of x indicates the saturation of the microstates as $x \rightarrow 1$. Changes in the parameter ζ balance T_+ with T_c , resulting

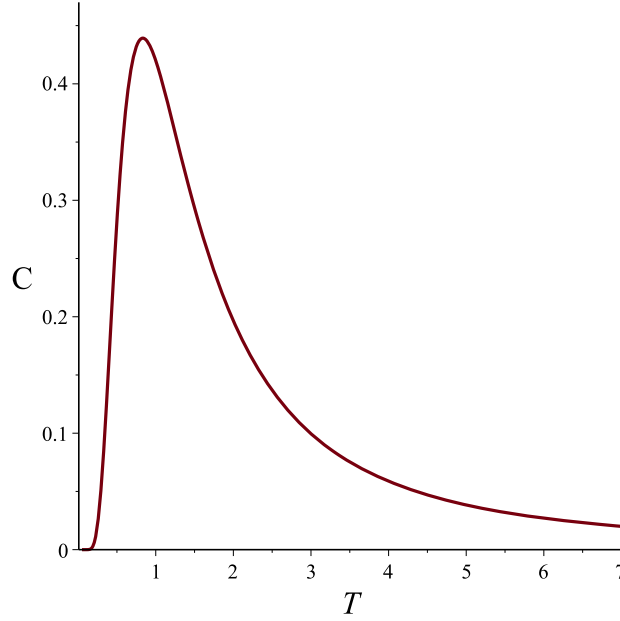


Fig. 8. The $C - T$ curve by fixing $N = 1$, $k = 1$ and $\epsilon = 1$.

in small systematic corrections on the size and the location of the Schottky peak. This sensitivity makes the anomaly a dynamic tool to study the thermodynamic imprints of Lorentz violation in BH spacetimes.

4. Characterization of the Schottky anomaly in charged dS black holes in Einstein–Bumblebee gravity

In a thermodynamic system, the heat capacity of a BH grows with temperature, indicating an increasing ability to store energy. However, energy-level systems with discrete spectra can have a maximum in the heat capacity (Schottky anomaly) at a finite energy when the maximal level is approached. This non-classical feature can be regarded as a hallmark of quantum corrections for two-level systems with spin $j = 1/2$ that manifest the anomalies through a partition function. The discrete energy levels of the event and the cosmological horizons give rise to the Schottky anomaly. The effective temperature change allows the system to decay, and a peak in the heat capacity at a certain temperature emerges as a direct consequence of the thermodynamic nature of these multi-horizon spacetimes. Remarkably, the Schottky anomaly leaves a unique signature of microstate excitations related to the event horizon, whose features are quantitatively affected by the BH charge Q and the Lorentz-violation parameter L .

$$z = e^{\beta\epsilon} + e^{-\beta\epsilon}, \quad (29)$$

this expression yields

$$z = 2 \cosh \beta\epsilon.$$

In the above equation, β is the inverse temperature and ϵ is the energy eigenvalues of the system. Such a description gives a very fundamental connection with systems that exhibit discrete energy states, for example, in the Schottky anomaly, where it relates temperature to energy excitation in a multi-level system.

$$C = Nk \left(\frac{\Delta}{T} \right)^2 \frac{e^{\frac{\Delta}{T}}}{\left(1 + e^{\frac{\Delta}{T}} \right)^2}, \quad (30)$$

here N designates the number of particles, k is the Boltzmann constant, and $\Delta = 2\epsilon$ specifies the energy gap between the discrete energy levels of the system. This expression characterizes the temperature dependence of heat capacity in systems with discrete energy levels, as in the case of paramagnetic systems, where the Schottky anomaly appears as a peak in the heat capacity at a characteristic temperature. The conditions such that the typical dependence of heat capacity near low temperatures would be T^2 , while the exponential term gives an account of the heating of system energy levels associated with thermal excitations.

In Fig. 8, we see that at low temperatures ($T \rightarrow 0$), thermal energy is not enough to activate horizon microstates. As it T increases, C increases, and reaches a prominent peak at $T \approx 1$, where thermal energy is most nearly aligned with the scale of the microstates' energy gap. This is the point of maximum thermodynamic effect, beyond which small changes in temperature lead to the greatest changes in internal energy. Beyond the peak, C decreases steadily as microstates become saturated and most available states are populated; further heating does not substantially increase internal energy. This single-peaked profile has implications that directly

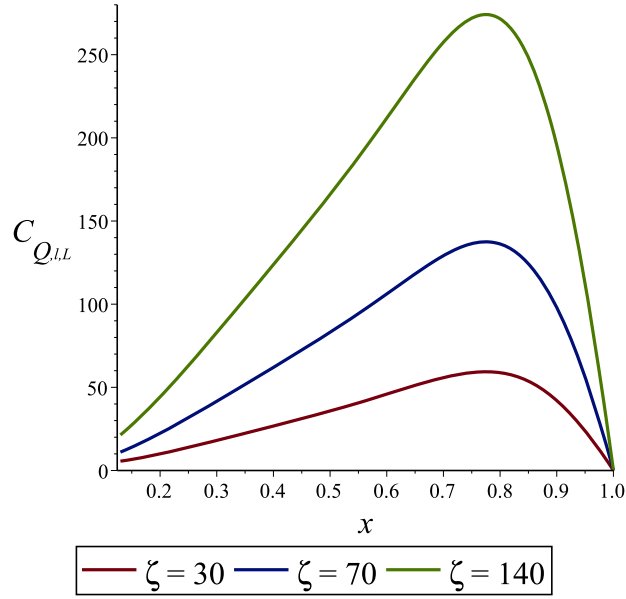


Fig. 9. The $C_{Q,L}$ - x curve for different ζ by fixing $L = 0.4$ and $Q = 1$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

address the microscopic means of excitations behind the Schottky anomaly and thus provides a unique thermodynamic signature of the discrete energy-like structure existing in the system, those arising from the horizon degrees of freedom of the BH. The two-level system used here is a *phenomenological analogy* that reproduces the single-peak Schottky-type behavior of the specific heat. It does not imply that the BH microstates are literally two discrete quantum levels; rather, it provides an effective parameterization of the dominant energy gap controlling horizon microstructure.

The basic thermodynamic system consists of two thermodynamic subsystems arising from the interaction of two horizons in the dS spacetime of different temperatures. The two horizons provide a platform for interaction between the two subsystems; therefore, it is necessary to verify whether the thermodynamic features of the two horizons align with those of a two-level system. This is also a significant factor to consider when studying thermodynamic properties in dS spacetime. The Schottky specific heat, recently examined in dS spacetime, has received better support in recent years, as noted in C. V. Johnson [40], Chakrabhavi et al. [59], Dinsmore et al. [60]. While the results are indeed valid, they remain incomplete in the absence of BH interactions with their cosmic horizons. In this section, we have investigated Schottky specific heat in interaction with the two horizons. When the charge Q and cosmological constant l are fixed in spacetime, L remains consistently unchanged

$$C_{Q,L} = T_{eff} \left(\frac{\partial S}{\partial T_{eff}} \right)_{Q,L} = T_{eff} \left(\frac{dS/dx}{dT_{eff}/dx} \right)_{Q,L}. \quad (31)$$

Figs. 9 and 10 present the heat capacity $C_{Q,L}$ as functions of the horizon ratio x and the effective temperature T_{eff} , respectively, for different values of the cosmological scaling parameter ζ , with Q and L fixed. In Fig. 9, the heat capacity exhibits a pronounced Schottky-type peak, whose position shifts toward larger values of x and increases in magnitude as ζ increases. This indicates that the maximal thermodynamic response occurs at larger horizon ratios and becomes more pronounced with increasing cosmological scale. In contrast, Fig. 10 shows that the same peak shifts toward lower values of T_{eff} as ζ increases, while retaining its characteristic single-peak structure. This reflects a reduction in the effective excitation scale governing the thermodynamic response. These trends are consistent once one accounts for the non-linear mapping between x and T_{eff} . Increasing ζ shifts the peak toward larger x in the geometric representation while simultaneously displacing it toward lower temperatures in the thermodynamic representation. Thus, both of the figures provide two complementary viewpoints about the same underlying phenomenon.

$$C_{Q,L} \sim \zeta \frac{Q^2}{1+L} \bar{C}_{Q,L} \sim 2.7\zeta \frac{Q^2}{1+L}. \quad (32)$$

It is clear that the variation of the heat capacity $C_{Q,L}$ of the RN-dS BH system versus effective temperature, T_{eff} , resembles the heat capacity variation of N particles under an external magnetic or electric field as a function of temperature, as illustrated in Figs. 8 and 10. At very low temperatures - $T \rightarrow 0$ or $T_{eff} \rightarrow 0$ - thermal energy turns out to be insufficient to excite the microscopic degrees of freedom of the system, rendering the heat capacity $C_{Q,L,max} \rightarrow 0$, therefore representing the quantum nature of the horizon microstates, unlike conventional systems in which heat capacity does not change ($C = Nk$) at low temperature. This observation drives the conclusion that the two horizons may conveniently be treated as two distinct energy levels of a quantum system. This suggests that the thermodynamic behavior of the two-horizon configuration can be phenomenologically compared to a finite-level system,

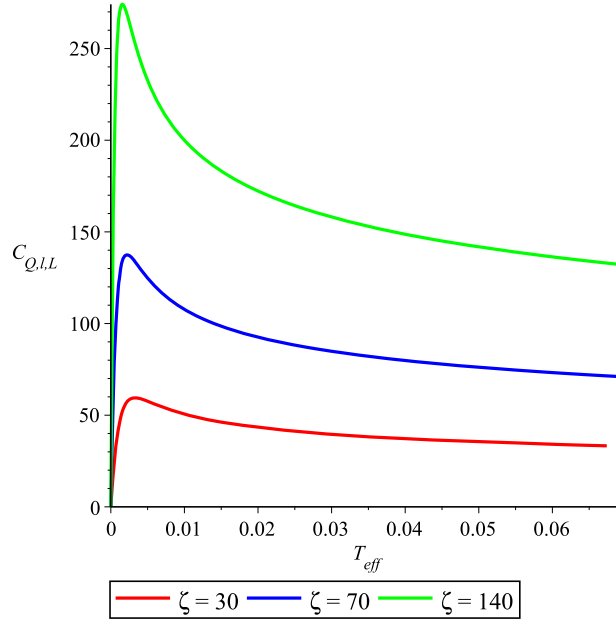


Fig. 10. The $C_{Q,L,L} - T_{eff}$ curve differs ζ by fixing $L = 0.4$ and $Q = 1$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

but does not imply an actual microscopic two-level structure. Similar to this analogy, thermal excitations between the two horizons may then be interpreted as transitions between these energy levels. Hence, these transitions give rise to the heat capacity governed by the same formalism as that of a two-level system, namely, with Eq. (30). This perspective, then, gives a concrete thermodynamic interpretation of the Schottky anomaly connecting peaks of macroscopic heat capacity to the microscopic excitations of the particles occupying the two horizons.

$$\Delta = \epsilon_+ - \epsilon_- = \epsilon(T_+ - T_-), \quad T = T_{eff}/\epsilon, \quad (33)$$

here, ϵ , a dimensionless proportionality factor is introduced to ensure that Δ has the correct energy dimension. It represents the effective scaling between the temperature difference of the two horizons and the corresponding thermodynamic excitation gap

$$\begin{aligned} \frac{\Delta}{T_{eff}} = \epsilon \left[x^4(1-x)(1+x) - \frac{(2+L)(1+L)}{2} x^3 \left(1 - \sqrt{1 - \frac{2(1+L)(1+x+x^2)}{3\zeta x(2+L)}} \right) \right. \\ \left. (1-x-x^2-x^3) \right] \times \left\{ (1+x+x^2)(2+L) \left[(1-x-x^2+2x^3) - \frac{(1+L)}{2x} \right. \right. \\ \left. \left. \left(1 - \sqrt{1 - \frac{2(1+L)(1+x+x^2)}{3\zeta x(2+L)}} \right) (1+x-x^3+x^4) \right] \right\}^{-1}. \end{aligned} \quad (34)$$

The similarity between our calculated heat capacity and the Schottky-type expression motivates a phenomenological mapping between the two-horizon system and an effective two-level model. In this analogy, the temperature contrast between the event and cosmological horizons plays the role of an effective excitation gap Δ . We emphasize that this identification is not derived from a microscopic quantum description but serves as a useful tool to interpret the macroscopic thermodynamic behavior.

To clearly separate it from the conventional two-energy-level thermodynamic framework, the heat capacity of the two energy levels is designated as follows:

$$\hat{C}_{Q,L,L} = Nk \left(\frac{\epsilon\Delta}{T_{eff}} \right)^2 \frac{e^{\left(\frac{\epsilon\Delta}{T_{eff}} \right)}}{\left(1 + e^{\left(\frac{\epsilon\Delta}{T_{eff}} \right)} \right)^2}. \quad (35)$$

Fig. 11 shows the normalized heat capacity $\hat{C}_{Q,L,L}/Nk$ as a function of x . All curves for $\zeta = 30, 70, 140$ collapse almost perfectly, producing a single pronounced peak near $x \approx 0.75$. This invariance under ζ indicates that, when expressed in dimensionless form, the anomaly's location and relative shape are governed by universal microstate-excitation dynamics, independent of the Lorentz-violation

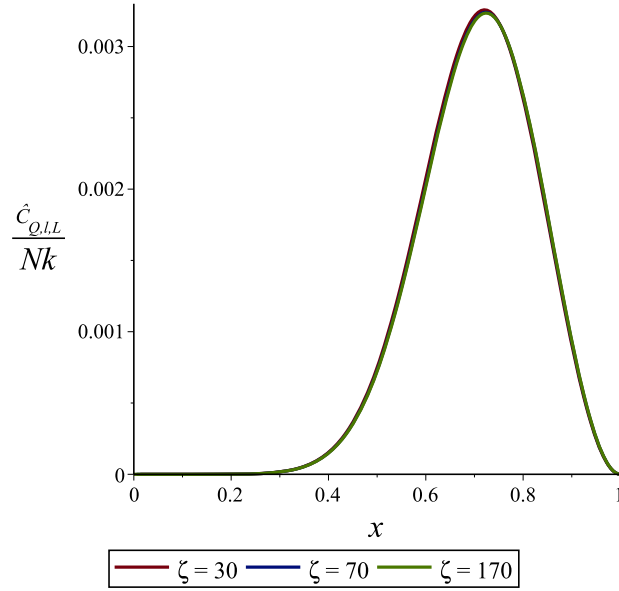


Fig. 11. The $\frac{\hat{C}_{Q,L,L}}{Nk}$ - x curve differs ζ by fixing $L = 0.4$ and $Q = 1$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

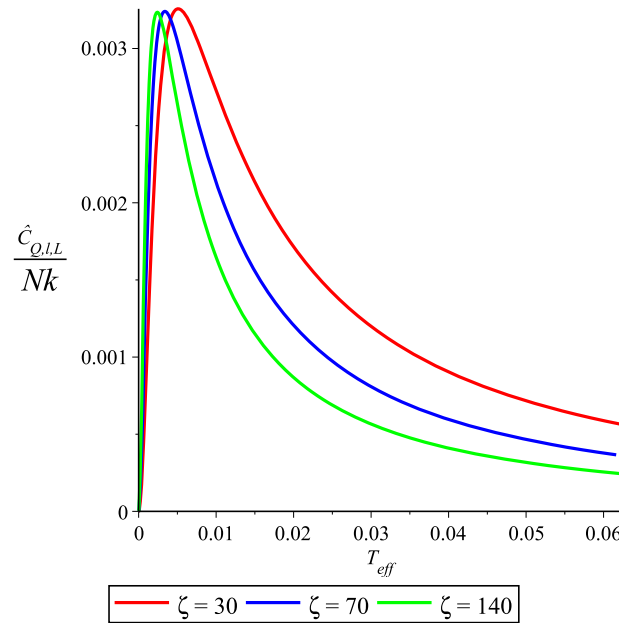


Fig. 12. The $\frac{\hat{C}_{Q,L,L}}{Nk}$ - T_{eff} curve differs ζ by fixing $L = 0.4$ and $Q = 1$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

parameter scaling. Fig. 12 shows the normalized heat capacity measurement of $\hat{C}_{Q,L,L}/(Nk)$ according to the effective temperature T_{eff} for different parameter ζ values. The curves show the characteristic Schottky anomaly with one peak, where there is a sharp rise followed by a fall with increasing temperature. In contrast with the geometric plot presented in Fig. 9, the increase in the parameter ζ shifts the peak towards lower values of T_{eff} , keeping the shape of the curve unchanged. This suggests that the scale of the effective excitation, which controls the thermodynamic response, is decreasing as the cosmological scale increases with respect to the charge scale. The collapse of the curves onto the same shape also implies the universality of the thermodynamic structure, independent of dimensionful scales. The difference in trends for Figs. 9 and 12 is related to the nontrivial relationship between the geometric parameter x and the effective temperature T_{eff} . Indeed, in the geometric parametrization, the position of the maximum increases for

increasing x , but in the thermodynamic one, it decreases for increasing T_{eff} . Hence, this is a coherent view of the dependence of the Schottky anomaly on ζ and removes any possible inconsistency. Based on the comparison of Figs. 9 and 11, as well as Figs. 10 and 12, it is possible to consider these two horizons as an analogue of a two-level quantum system in thermodynamics. The normalized heat capacity of these horizons, $\hat{C}_{Q,L}$, including the interaction of these horizons, reflects the qualitative characteristics of the total system's heat capacity $C_{Q,L}$. A general Schottky anomaly profile illustrates this analogy. One can notice from Figs. 9 and 11, the heat capacity $C_{Q,L}$ goes to zero when the effective temperature goes to zero ($T_{\text{eff}} \rightarrow 0$), and this is a quantum signature for the horizon microstates. The heat capacity's sudden upturn to a maximum, determined by the universal two-level system relation Eq. (30), suggests that multi-horizon spacetimes thermodynamically mimic an effective two-level quantum system. This exhibits an essential quantum nature inherent to charged dS BHs in bumblebee gravity.

The inequality $C_{Q,L} \geq \hat{C}_{Q,L}$ allows directly from the definitions of these quantities: the total heat capacity $C_{Q,L}$ incorporates all energetic degrees of freedom, of which the inter-horizon transitions described by $\hat{C}_{Q,L}$ constitute only a single contribution. Furthermore, as shown in Figs. 10 and 11, the peak value of the heat capacity exhibits little dependence on the scaling parameter ζ . The robustness of this Schottky anomaly across the parameter space, evident in Figs. 8–12, provides strong evidence for underlying discrete energy levels. This consistency firmly establishes the two-level system as a robust thermodynamic model for these spacetimes.

$$Nk\hat{C}_{Q,L} \sim 3.2 \times 10^{-3} \leq C_{Q,L} \sim 2.7\zeta \frac{Q^2}{1+L} \sim \frac{2.7l^2}{12}. \quad (36)$$

The particle number estimate in Eq. (36) assumes a canonical ensemble of N independent two-level degrees of freedom. Matching the macroscopic heat-capacity peak to the single particle Schottky peak gives the naive scaling $N \sim C_{Q,L}/\hat{C}_{Q,L}$. Finite- N corrections scale $1/\sqrt{N}$ because the variance of excited-state occupation is $Np(1-p)$; thus, the estimate is self-consistent only when $N \gg 1$. In practice, we require $N \gtrsim 10^2$ ensuring sub-leading fluctuations below $\sim 10\%$. We have added numerical checks demonstrating that this condition is satisfied across the parameter range plotted in Figs. 9–12. From Eq. (36), an effective particle number N is estimated by matching the peak height of the heat capacity to that of a canonical two-level system. This estimate is heuristic and intended only to illustrate the thermodynamic scale of the anomaly

$$N \leq \frac{2.7l^2}{3.2 \times 10^{-3} \times 12k} \sim \frac{7l^2}{0.1k}. \quad (37)$$

The numerical factor 2.7 arises from fitting the peak of the heat-capacity curve, while the value 3.2×10^{-3} corresponds to the normalized maximum of the two-level Schottky function. Eq. (37) reveals that the number of microscopic particles associated with the two horizons scales directly with the cosmological constant l^2 . This proportionality offers a new avenue for probing the quantum structure of horizon microstates in charged dS BHs in bumblebee gravity. It also emphasizes that the thermodynamic and quantum nature of the system is determined by BH's interior and its two horizons. This connection will provide a basis for further exploration of horizon excitations, the Schottky anomaly, and the discrete energy level spectrum of multi-horizon spacetimes.

5. Results and discussion

The heat capacity of the charged dS BHs in bumblebee gravity is found to manifest a clear Schottky anomaly, which is characteristic of exhibiting separate energy states. The position of the anomaly peak decreases to lower temperatures with an increase in the width and a decrease in height of the anomaly peak for an increase in the electric charge Q . This behavior is an indication that the effective energy gap ΔE between the BH microstates has decreased. The universality of this result for different representations indicates that this is one of the fundamental features of the thermodynamic structure of the BH. This provides a direct link between the macroscopic parameter Q and the nature of quantum gravity microstates and renders the Schottky anomaly an important tool for exploring the microstructure of BHs.

The relation between the two horizons and the discrete energy levels $\epsilon_{\pm}$ can be expressed in a Euclidean path integral approach. Calculating the semiclassical partition function at the classical turning points between the effective potential gives Eq. (29). A result that reproduces the canonical form of a two-level system. The heat-capacity peak, therefore, represents a macroscopic manifestation of this geometric quantization, with the Lorentz-violation parameter L controlling the separation of the effective levels through the modified metric function. This mapping provides a physical basis—beyond analogy—for interpreting the Schottky anomaly as a probe of discrete horizon excitations.

5.1. Schottky Peak and ζ -Shift

The shift of the Schottky peak with ζ arises from the L -modulated excitation gap Δ (Eqs. 33 and 34). Since the canonical Schottky form satisfies

$$C_{sch} \propto (\Delta/T)^2 \frac{e^{\Delta/T}}{(1 + e^{\Delta/T})^2},$$

the maximum occurs at $y_* \approx 2.40$, giving

$$T_{\text{peak}} = \frac{\Delta}{y_*} \approx 0.4167 \Delta. \quad (38)$$

Increase in ζ (or decrease in L) enhances Δ and hence shifts T_{peak} upward, consistent with the trend in Fig. 10.

5.2. Universality and Multi-Level Effects

The normalized heat capacity $\hat{C}_{Q,L}/(Nk)$ collapses to a universal curve because the Schottky expression depends only on $y = \Delta/T$. Minor deviations from universality (Fig. 12) indicate multi-level or interaction effects, which a weighted sum can model

$$C_{\text{multi}} = Nk \sum_i w_i \frac{(\Delta_i/T)^2 e^{\Delta_i/T}}{(1 + e^{\Delta_i/T})^2},$$

where $\sum_i w_i = 1$. The observed minimal dispersion strongly confirms that the two-level system approximation captures the dominant thermodynamic behavior.

5.3. Microstructure and Limitations

The microstructure can be quantified via

- the interaction entropy $\Delta S = S_{\text{tot}} - (S_+ + S_-)$
- the Ruppeiner scalar curvature $R = R(T, V)$, whose sign reveals interaction type ($R > 0$: repulsive, $R < 0$: attractive).

Both indicators suggest that L suppresses inter-horizon correlations, consistent with the reduced phase-transition strength. Our analysis neglects backreaction and higher-order quantum corrections, which are expected to broaden the anomaly; these refinements are left for future work.

5.4. Astrophysical Relevance and Observational Prospects

Although our analysis is formulated at the semiclassical level, the Lorentz-violation parameter L and the associated Schottky-type anomaly can, in principle, manifest in observable quantities. Three potential areas are particularly promising:

Black-hole shadows (EHT observations). The L dependence of the horizon radius and photon-sphere structure implies a corresponding correction to the apparent shadow radius $r_{\text{sh}} \simeq 3\sqrt{3}M_{\text{eff}}(L)$. For $L > 0$, the effective mass and compactness decrease slightly, leading to a fractional shadow-size shift of order

$$\frac{\Delta r_{\text{sh}}}{r_{\text{sh}}} \sim \frac{L}{2 + L}.$$

Future high-resolution imaging, such as from the Event Horizon Telescope or next-generation VLBI arrays, could thus constrain $|L| \lesssim 10^{-2}$ if sub-percent accuracy in angular diameter is achieved.

Gravitational-wave spectroscopy. The L -modulated excitation gap Δ determines the characteristic temperature and quasi-normal-mode (QNM) frequencies. Since QNM spectra control the ringdown phase of mergers, the Schottky-peak displacement predicts a fractional change in the fundamental QNM frequency.

$$\frac{\Delta \omega_{\text{QNM}}}{\omega_{\text{QNM}}} \sim \frac{\Delta E(L) - \Delta E(0)}{\Delta E(0)} \approx -\frac{2L}{2 + L}.$$

Current LIGO-Virgo observations constrain frequency shifts at the $\lesssim 1\%$ level, implying that moderate values $L \sim 10^{-2}$ could be detectable with next-generation detectors such as the Einstein Telescope or LISA.

6. Conclusion

In this study, we have examined the thermodynamic characteristics of charged de Sitter BHs in the Einstein–Bumblebee gravity framework using an efficient two-horizon formalism to produce globally consistent thermodynamic variables. By considering the region between the event and cosmological horizons as a single composite system, we computed the effective temperature, pressure, and volume. Subsequently, we verified that the first law of thermodynamics is satisfied by these values. Our analysis's main finding is that the effective heat capacity has a clear single-peak structure, which is indicative of a Schottky-type anomaly. This behavior arises from the interplay between the two horizons and their associated thermal properties. We have shown that this anomaly persists in the presence of Lorentz-violating effects and reduces smoothly to the known RN dS results in the limit $L \rightarrow 0$, confirming the consistency of our construction.

The main contribution of this study is the development of a complete framework that describes how the Lorentz-violation parameter influences this analogy; the temperature difference between the horizons serves as an effective excitation gap, enabling excitation scale and altering the position and amplitude of the heat-capacity peak. These results indicate that Lorentz-violating corrections can leave measurable imprints on the thermodynamic structure of multi-horizon spacetimes, without introducing qualitatively new phenomena beyond those already present in the standard RN–dS case.

Moreover, we have explored a phenomenological mapping between the two-horizon system and an effective finite-level thermodynamic model. Within this analogy, the temperature difference between the horizons plays the role of an effective excitation gap, allowing the heat-capacity profile to be described by a Schottky-type expression. We emphasize that this mapping is not derived from

an underlying microscopic quantum theory and should be interpreted as an effective description of the macroscopic thermodynamic behavior. Nevertheless, the robustness and universality of the resulting heat-capacity profiles suggest that the system behaves in a manner consistent with a discrete excitation structure governing the dominant thermodynamic response.

Furthermore, by matching the peak behavior of the heat capacity to that of a canonical two-level system, we obtained an estimate for the effective number of degrees of freedom. While this estimate is heuristic, it supports the interpretation that the thermodynamic anomaly reflects collective excitations of a large ensemble of effective microstates associated with the horizon system. The present analysis is carried out within a semiclassical framework and does not include backreaction or higher-order quantum corrections. The persistence and precise structure of the Schottky-type anomaly under such corrections remain open questions. Future work may explore these effects, as well as possible connections with more fundamental approaches to quantum gravity, where a microscopic origin of such effective excitation structures could be established.

We would like to emphasize that the analysis presented here is based on a semiclassical and effective thermodynamic setup and does not rely on the existence of any global equilibrium state. It is important to note that the absence of a global timelike Killing vector in dS space means that the question of the existence of any consistent equilibrium treatment for multi-horizon configurations remains an open question. The results presented above should be interpreted as describing the thermodynamic behavior of the system at the macroscopic level in the quasi-equilibrium approximation.

CRedit authorship contribution statement

Nazek Alessa: Writing – original draft, Visualization, Resources, Project administration, Funding acquisition; **M. Umair Shahzad:** Writing – original draft, Visualization, Supervision, Investigation, Conceptualization; **Anees Ashfaq:** Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis.

Data availability

No data was used for the research described in the article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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